

Master DSC/MLDM - First year

Introduction to Artificial Intelligence

First exam - November 2025

Take the things you need with you. Leave your smartphone, calculator and computer in your bag and place your bag at the front of the examination room.

Time allocated: 3h - No documents allowed.

TAKE CARE: any cheating will be severely punished and will lead to a formal complaint to the disciplinary council of the university.

1 Truth table ($\simeq$ 2 points)

We consider a universe with two constants a and b , one predicate r of arity 2, and two predicates p and q of arity 1. Using the truth table method, as we did in class, say whether the two following first order sentences are unsatisfiable, valid or contingent.

1. $\exists Y. \forall X. r(X, Y) \Rightarrow \forall X. \exists Y. r(X, Y)$
2. $\exists X. (p(X) \wedge \neg q(X)) \wedge \forall X. (p(X) \Rightarrow q(X))$

Note 1: Using any other technique will result in a score of 0 for this exercise.

Note 2: For that question, you should present your table in the same way as the example below showing the truthtable of $(x \vee y) \wedge \neg x$:

x	y	$(x \vee y) \wedge \neg x$				
T	T	T	T	T	F	F
T	F	T	T	F	F	F
F	T	F	T	T	T	T
F	F	F	F	F	F	F

2 Problem modeling and solving in PL ($\simeq$ 5 points)

Here are some facts about a simple world taking place in a computer science class:

When the Wi-Fi stops working, the students get nervous, they don't have access to chatGPT and the teacher gets angry. If the students get nervous then the server crashes. When the teacher gets angry, he gives a surprise quiz. If the teacher gives a surprise quiz, everyone is shouting at David. If everyone is shouting at David and David panics, then he throws his computer into the classroom. When the teacher gives a surprise quiz, David panics and drinks too much coffee. When David drinks too much coffee, he talks loudly and disturbs Thomas. When David disturbs Thomas and throws his computer into the classroom, Thomas loses focus and forgets to save his code. If Thomas loses focus and the teacher gives a surprise quiz, the project is delayed. If the project is delayed, the teacher gets angrier. When the teacher gets angrier and the server crashes and David talks loudly, the Wi-Fi technician is called. If the Wi-Fi technician is called, everybody feels better. If everybody feels better, David and Thomas are good friends again. The Wi-Fi stops working.

You are asked to:

1. Choose proposition constants wisely (list all your constants on your exam sheet).
2. Convert these English sentences into propositional logic sentences (list all those sentences on your exam sheet).
3. Convert these proposition logic sentences into CNF (list all your clauses on your exam sheet).
4. Use resolution reasoning in proposition logic to prove that, in this world: *David drinks too much coffee, Thomas forgets to save his code and they are good friends again* (write all the steps of the reasoning on your exam sheet).

Note: Modelling this universe using first-order logic will result in a score of 0 for this exercise.

3 Validity, unsatisfiability, contingency ($\simeq 5$ points)

Using resolution reasoning and the method studied during the course, say whether the following sentences are unsatisfiable, valid or contingent

1. $(p \Rightarrow q) \Rightarrow (\neg q \Rightarrow \neg p)$
2. $((p \vee \neg q) \wedge (r \vee q)) \Rightarrow (p \vee r)$
3. $\forall X.(p(X) \Rightarrow q(X)) \wedge \exists X.(p(X) \wedge \neg q(X))$
4. $\forall X.p(X) \Rightarrow \exists X.p(X)$
5. $\forall X.\forall Y.((lt(X, Y) \Rightarrow \neg lt(Y, X)) \vee \exists Z.\forall X.(lt(X, Y) \Rightarrow \forall X.lt(Z, X)))$

Note 1: you must write down all the details of the conversion of each formula to CNF.

Note 2: Using any technique other than resolution reasoning, as seen in class, will result in a score of 0 for this exercise.

4 Resolution principle ($\simeq 3$ points)

For each pair of clauses below, say whether the resolution principle can be applied. If yes, explain the steps you follow and give the resolvent(s), if no, explain why.

1. $\{p1(X), p2(X, a, f1(X))\}$ and $\{\neg p2(b, Y, f1(b)), p3(Y)\}$
2. $\{\neg p1(X, f1(Y), k), \neg p1(f2(a, b), f1(c), W), \neg p2(W)\}$ and $\{p1(A, f1(B), D), p3(A, B)\}$
3. $\{p1(X, a, Y), p2(Y)\}$ and $\{p3(X), \neg p1(b, X, Z)\}$
4. $\{p1(a, f1(X, f2(Y), Z), b, T), p2(X, Y), p1(X, f1(a, T, c), Y, f2(Y))\}$ and $\{\neg p1(A, f1(C, D, E), b, F), p3(A, F)\}$
5. $\{p1(A), \neg p2(b)\}$ and $\{\neg p1(a), p2(B)\}$
6. $\{p2(A), p1(A, B), p3(B), p1(a, C), p4(C), p1(A, b)\}$ and $\{\neg p1(A, b), \neg p1(a, B), p2(A), p5(B)\}$

5 Problem modeling and solving in FOL ($\simeq 5$ points)

Below is information about a simple world, for any person P1, P2 and P3.

- If a person P1 is richer than another person P2, and they live in the same place then P1 pays more taxes than P2.
- If a person P1 is richer than another person P2, and P1 lives in town and P2 lives in the countryside, then P1 is smarter than P2.
- If a person P1 is richer than a person P3 that is richer than a person P2, then P1 is richer than P2 (the relationship "richer than" is transitive).
- The relationship "younger than" is also transitive.
- If a person P1 is smarter than a person P2, or if P1 is younger than P2 and P1 is a student and P2 is a teacher, then P1 is happier than P2.
- If P1 pays more taxes than P2 then P1 is jealous of P2.
- If P1 is happier than P2 or P1 is jealous of P2, then P1 has feelings.
- Bess and Dana live in the same place.
- John is a student.
- John is younger than Mary.
- Suzy is a teacher.
- Cody lives in town.
- Cody is richer than Dana.
- Dana lives in the countryside.
- Bess is richer than Cody.
- Mary is younger than Suzy.

You are asked to:

1. Convert these English sentences into first-order logic sentences.
2. Convert these first-order logic sentences into CNF (you can directly write the clauses without showing the steps of conversion).
3. Use resolution reasoning in first-order logic, to find all the persons that have feelings. (write all the steps of the reasoning on your exam sheet)